



3D-PATH: A Hierarchy LUT Processing-in-memory Accelerator with Thermal-aware Hybrid Bonding Integration

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Abstract

LUT-based processing-in-memory (PIM) architectures enable general-purpose in-situ computing by retrieving precomputed results. However, they suffer from limited computing precision, redundancy, and high latency of off-table access. To address these challenges, we present 3D-PATH, a novel PIM architecture that employs 3D hybrid bonding to integrate a DRAM-LUT, enhancing system capacity and reducing access latency. To further optimize efficiency, 3D-PATH introduces a hierarchical fast-LUT design that reduces storage redundancy and accelerates computation. Additionally, 3D-PATH extends computing precision by efficiently supporting floating-point operations via representation transformation and parallel interleaving banks. While hybrid bonding offers significant benefits, it induces heat dissipation challenges. To address this, we implement thermal-aware hardware that ensures the DRAM Die temperature maintains below the threshold of 85°C. Evaluations on arithmetic and AI workloads demonstrate that 3D-PATH achieves up to 12.68× higher throughput than GPUs and 2.27-7.54× over prior LUT-PIMs, while delivering a 12.24× improvement in floating-point energy efficiency over GPU and 2.13× over a 3D baseline.

Keywords

3D Architecture, Hybrid Bonding, LUT, Processing-in-memory

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1 Introduction

Advances in hardware have fueled the rapid growth of applications like AI and IoT, which increasingly demand efficient processing of high-dimensional contextual data. However, these workloads

place stringent requirements on hardware, as extensive data movement leads to significant power consumption and latency bottlenecks [1, 9, 56]. Processing-in-memory (PIM) technology offers a solution by enabling in-situ computing, thus greatly reducing or even eliminating data movement and enhancing power efficiency [5, 10, 11, 15, 19, 25, 40, 43, 48, 62, 63, 68, 74, 75]. Nonetheless, PIM architectures face a trade-off between functionality and hardware overhead [17]. Specifically, certain PIM designs efficiently support basic logic operations, such as Boolean AND and OR, within the memory array. In contrast, more complex operations, including multiplication and non-linear functions, result in increased latency. Some PIM designs incorporate function-complete units at the bank level or within the memory controller to preserve the integrity of the memory array. However, this strategy incurs significant area overhead, as DRAM technology restricts the number of available metal layers for placement and routing. For instance, HBM-PIM implementations can lead to a 50% area overhead when integrating computing units between DRAM banks[39].

Recently, lookup table (LUT)-based PIM designs have been proposed to resolve limitations related to functionality and area overhead [17, 20, 29, 59, 66, 69, 77]. This approach allows for the retrieval of results for complex operations directly from a memory array. Theoretically, LUT-based PIM supports general-purpose operations without the need for intricate computing units, thereby mitigating related overhead.

However, LUT-based PIM faces several challenges. (1) Limited computing precision: current LUT-PIM designs primarily accommodate integer operations, whereas emerging applications increasingly demand high-precision computations, particularly involving floating-point operations. These operations introduce additional complexities, such as exponent alignment and the separate handling of mantissa and exponent, leading to non-negligible latency. (2) Overlooked storage redundancy: Prior LUT designs fail to account for storage redundancy caused by sparsity. Specifically, these designs can hardly identify or eliminate redundant elements of a sparse matrix, leading to unnecessary storage and corresponding computation. (3) Inefficient ‘off-table’ operation: LUT designs suffer from significant latency/power between the DRAM and the logic units due to lengthy wiring and parasitic capacitance. This issue becomes particularly critical in LUT-PIM when a single LUT cannot fulfill all required tasks or requires updating, necessitating pre- or post-processing by ‘off-table’ units and associated data movement.



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This work addresses the limitations of prior LUT-PIM designs by introducing a **3D PIM Accelerator with Thermal-aware Hybrid bonding integration (3D-PATH)**. 3D hybrid bonding integration stacks DRAM on top of the logic unit and interconnects them using ultra-high-density copper pillars, achieving high parallelism and bandwidth. This 3D technique motivates a LUT-based PIM design, where the key idea is to **transform the high bandwidth of 3D integration into high LUT throughput**. Additionally, hybrid bonding decouples the fabrication of logic and DRAM Dies, allowing the logic Die to use advanced technology while preserving the DRAM Die's integrity. In this design, the stacked DRAM functions as a parallel, high-capacity but slower LUT to store the dominant workload, while the logic Die incorporates a fast-LUT to store key indices. This hierarchical LUT exploits data redundancy by first identifying fast-LUT entries to determine valid DRAM table indices, and then activating the DRAM LUTs to retrieve computation results. Furthermore, 3D-PATH facilitates full precision floating-point operation with a multiplier-free and alignment-free design.

The hybrid bonding integration shortens the distance between DRAM and logic units, but it also introduces significant thermal challenges. Specifically, the logic Die is sandwiched between the DRAM and the package, preventing direct contact with the cooling system. Elevated temperatures can reduce the operating frequency and data holding period. To address these thermal concerns, we propose a hotspot-aware 3D hardware, which dynamically tunes the functioning state, mitigating excessive thermal buildup. The contributions of this work are:

- We present one hybrid bonding-based accelerator design choice, **3D-PATH**. This design capitalizes on the high bandwidth and storage density of 3D integration to achieve a high-throughput and area-efficient LUT-based PIM.
- We propose a **sparse-aware** hierarchical LUT that preserves the integrity of stacked DRAM as parallel, high-capacity LUTs while integrating fast-LUTs in the logic Die. This combination efficiently detects sparsity, reducing redundant storage and computation.
- We introduce a data format transformation mechanism that enables multiplier-free **floating point** operation, adapting to diverse computational requirements. Additionally, an interleaved mapping facilitates alignment-free computations.
- To address the thermal challenges of hybrid bonding integration, we design a **hotspot-aware** hardware incorporating low-toggle-rate peripherals and a self-throttling LUT. This mechanism dynamically adjusts the system's operational state to mitigate heat buildup.
- We thoroughly evaluate the thermal issue and the integration overhead of 3D-PATH, demonstrating its efficiency through circuit and architecture performance analysis.

2 Background

2.1 LUT-based Processing-in-memory Design

PIM has emerged as a promising solution to mitigate the beachfront issue by enabling computation directly within the memory array [18, 23, 35, 39, 63]. However, prior PIM designs face a trade-off between functionality and performance. Coarse-grained designs provide broader functionality but suffer from lower parallelism and

larger compute unit sizes, with metal-layer constraints in DRAM further exacerbating area overhead. In contrast, fine-grained designs offer higher computational efficiency but are constrained in functionality.

LUT has long been employed in FPGA and ASIC designs for reconfigurable computing [31, 38, 46, 52, 64, 65, 67]. It transforms computations into memory lookups, where precomputed results are stored, and input data retrieves matching entries. Recent works have adapted this concept to PIM architectures and addressed the limitations of conventional PIM [11, 17, 29, 55, 59]. This approach eliminates complex computational logic, reducing area and energy overhead while enabling functionally complete logic. E.g., pLUTo [17] utilizes LUT to achieve 2.94× higher performance/area than Ambit. Despite its advantages, LUT-based PIM introduces several challenges:

Limited Precision: Traditional LUT-based PIM designs primarily support integer computations, simplifying implementation but limiting applicability. Many modern applications require FP computation to handle outliers and minimize quantization error. But FP involves exponent alignment and mantissa processing steps that do not align with LUT-based memory reads.

Redundant Data: Many LUT entries correspond to sparse data that do not contribute to the final result. For example, in matrix multiplication, conventional LUTs perform exhaustive searches over all input combinations, missing the opportunity to skip sparse data [54, 72].

Inefficient Access: LUT-based PIM faces challenges when the stored data becomes outdated or when a single LUT is insufficient to complete the required task. In such cases, the LUT needs to be updated or relies on off-memory units for pre-/post-processing. This incurs time/power-consuming off-memory data movement.

In summary, an effective LUT-based PIM design is expected to support high-precision computations, optimize data-aware operations to reduce redundancy, and enable fast off-table access.

2.2 Hybrid Bonding Technique

2.2.1 Hybrid Bonding Integration. Hybrid bonding has emerged as a pivotal technology for overcoming memory bandwidth limitations [51, 73]. As illustrated in Fig. 1, this architecture enables independent fabrication of DRAM and logic Die, followed by a vertical integration: The DRAM Die is stacked on the logic Die, with their top metal layers interconnected via dense copper pillars. These sub-3 μm pitch pillars achieve single-cycle data transmission between Dies. Crucially, the integration partitions DRAM into compact banks, each directly interfaced with dedicated logic units through pillars whose parallelism scales with pillar density (maximum 110,000/mm²), achieving > 16 TB/s bandwidth.

While the finest granularity of control is at the 2MB bank level, each DRAM bank can be subdivided into mats. During a read operation, command and address (C/A) signals are transmitted through the copper pillar, activating the target mat rows. These rows are read out to the bitlines (BLs), where a sense amplifier differentiates the small voltage signal using a cross-coupled latch, driving the bitline voltage rail-to-rail. The column address selects the data, which is subsequently read out to the logic unit through the pillar.

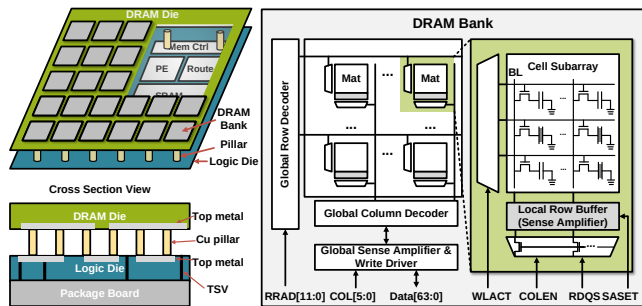


Figure 1: Overall Structure; Profiling view; DRAM array

2.2.2 Hybrid Bonding DRAM vs. Conventional Stacked DRAM. Stacked DRAM, such as HBM and HMC, have been proposed to enhance storage density and bandwidth, with PIM designs demonstrating high energy efficiency[4, 16, 21, 24, 41]. However, these architectures rely on Through-Silicon Vias (TSVs) to interconnect multiple Dies. TSVs penetrate the substrate, creating dead space where no devices or routing can be placed. The dense placement of TSV fragments the Die, impacting DRAM integrity. As a result, stacked DRAM architectures typically limit TSV counts below 1024, constraining bank-level parallelism (e.g., 16 parallel vaults in HMC). Additionally, the long TSVs introduce parasitic capacitance and high power consumption.

Hybrid bonding enables decoupled fabrication and then integrates the metal layers of logic and DRAM Dies using copper pillars, avoiding TSV-induced floorplan constraints. The bonding pitch is significantly smaller than TSV ($< 3 \mu\text{m}$ Copper pillar vs. $40\text{-}50 \mu\text{m}$ TSV [16]). This architecture enables **high bank-level parallelism**, substantially increasing bandwidth (e.g., 4096 parallel banks in this work). Hybrid bonding achieves significantly **lower I/O power** consumption, less than 0.3 pJ/b . While TSVs are still present, their role shifts to power delivery and off-chip I/O, greatly reducing the required density. Further overhead analysis is provided in Sec. 7.3.

3 Motivation: Hybrid Bonding-based LUT

The challenges inherent in LUT-based PIM necessitate the development of an innovative PIM architecture that offers higher storage density, reduced access latency, and lower power consumption. Such advancements would address the substantial storage demands of larger LUTs and minimize system stalls caused by data transfers between LUTs and off-table units.

Therefore, the insight of this work is to **translate the advantages of hybrid bonding technology into enhanced LUT performance**. The architecture leverages several key benefits of 3D integration: (1) The native bank-level parallelism maximizes lookup throughput. (2) The reduced access latency inherent to hybrid-bonded interfaces facilitates efficient LUT refresh operations and pre/post-processing. (3) The architecture innovation lies in the technology decoupling enabled by hybrid bonding. By implementing logic units and DRAM on separate wafers, the logic components are not constrained by limited DRAM metal layers. This flexibility

provides an opportunity to address fundamental limitations in conventional LUT-based PIM designs, particularly data format rigidity and storage redundancy.

We propose a hierarchical LUT architecture enabled by hybrid bonding. The DRAM Die serves as a high-capacity but **slow LUT** for general-purpose operations, while the logic Die integrates a **fast LUT** that acts as an index filter, forwarding only relevant queries to the DRAM LUT. This selective access reduces redundant lookups and minimizes storage overhead. Additional digital units support more complex operations, such as FP computation.

4 3D-PATH Architecture

This section has outlined the overall architecture of the 3D-PATH, which comprises two main components: the DRAM Die and the logic Die. The DRAM Die is partitioned at the bank level, with each bank functioning as a dedicated lookup table. The entry containing precomputed results, is then retrieved to the logic units through the copper pillar. The logic Die incorporates customized hardware, like a fast-LUT that accelerates lookup operations and reduces storage requirements. Finally, an example of the floating-point General Matrix Multiplication (GEMM) is discussed to illustrate its improvement.

4.1 Hybrid Bonding DRAM-LUT Organization

This work implements a DRAM-LUT that harnesses the high parallelism of hybrid bonding and the high burst capabilities of DRAM to enhance performance. Fig. 1 provides an overview of the DRAM organization within the hybrid bonding architecture. From a chip perspective, the DRAM Die consists of 4096 parallel banks, each connected to its corresponding logic unit via copper pillars, which serve as the datapath between two Dies. The number of copper pillars determines the I/O parallelism. By leveraging this parallelism, the hybrid bonding architecture significantly boosts the bandwidth, then translates into high-throughput LUT operations.

DRAM’s high burst capability enables efficient memory access by leveraging the disparity between row activation and column access latencies. Activating an entire row and loading it into the global buffer (tRCD = 17ns) incurs significantly higher latency than selecting partial data via the column address (tCCD = 2ns). To exploit this, table entries are allocated at the row level, with the column address determining the target entry within each row. Each DRAM bank consists of 4096 rows and 4096 columns, supporting a maximum burst size of 64 bits, corresponding to a table entry size of 64. More table entries can be allocated by reducing the width of each entry. Reducing the burst width to 32 bits increases the entry number to 128. Different rows within a DRAM bank can store different LUTs for various functions. During a lookup operation, the DRAM-LUT activates the row containing the required table according to the function type. The entire selected row is read through the BLs into the sense amplifier, after which the column address selects the target LUT entry based on the input combination, retrieving the precomputed result from the SA.

The 3D-PATH enables 4096 parallel 64-b DRAM lookup operations each cycle. The efficiency of the DRAM-LUT is closely tied to the available I/O parallelism, as the lookup operation is fundamentally a read operation. Hybrid bonding vertically stacks DRAM

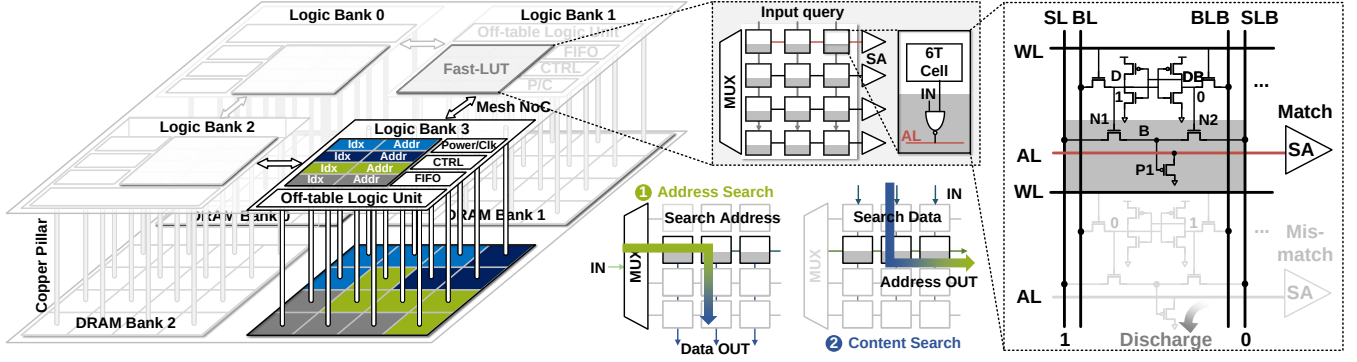


Figure 2: Hybrid bonding architecture (flipped) Top: Logic bank, Bottom: DRAM bank; and Bidirectional fast-LUT circuit design

and logic Dies, enabling a flexible I/O placement across the chip. This significantly enhances the throughput of lookup operations. Additionally, the fine-grained organization of DRAM banks ensures efficient storage of LUT entries, maximizing utilization.

4.2 Content Addressable Fast-LUT Design

Hybrid bonding decouples the fabrication of DRAM and logic Dies. This flexibility allows designing the logic unit independently of the DRAM's area and technology constraints. Therefore, the logic Die includes extra hardware to facilitate the operation of DRAM-LUT.

To accelerate lookup operations, we propose a fast-LUT in the logic Die to assist with DRAM-LUTs. Multiple tables are stored row-by-row in DRAM, the logic Die has to iteratively identify the target table during computation. To optimize this process, the logic Die integrates a fast-LUT, which serves as an abstraction of the DRAM-LUT, storing indices for each function. This design enables fast identification of the target table, significantly reducing the latency. The fast-LUT is based on the content addressable memory (CAM) to perform parallel content searches [7, 22, 28, 30, 32, 71]. During operation, the fast-LUT can compare the input query with the stored content to identify the target DRAM-LUT within 1 cycle.

To support efficient content searching, we implement a bidirectional fast-LUT design, as illustrated in Fig. 2. The CAM is based on a 9T array, where each cell consists of 6T for storage and three transistors (N1, N2, P1) for comparison. The input query is applied on SL/SLB. When the stored data matches the input key, node B maintains at high state, and P1 is shut off. When mismatches, the node B drops low, creating a discharge path from the accumulation line (AL) to ground. Therefore, only when the cell from entire row matches the input key, the ML could hold at high state. A current sense amplifier then quantizes the current value on AL to digital output, identifying the address of the matching entry.

In addition to its content-search capability, the design also supports an address search mode, where content can be searched by address. This is achieved using the original 6T SRAM cell: the address activates the wordline (WL), and the data stored in the cell is read out through the bitlines (BL/BLB). A sense amplifier then differentiates the values, allowing the design to perform a bidirectional search-either from address to content or from content to

address. By combining these two functionalities, we create a fast-LUT that efficiently locates the target table in DRAM and retrieves the address value of the corresponding DRAM-LUT.

4.3 Other Digital Peripherals

The logic Die integrates essential peripherals to complete pre/post-processing tasks. The Off-table Logic Unit (OLU) includes necessary function units, like a dual-adder tree for merging partial products from the LUT, as detailed in Sec. 5.3.1. It also incorporates a centralized memory controller to access 16 neighboring DRAM banks at the bank-level granularity. Inter-logic bank communication relies on the mesh NoC, which includes a 64-bit wide flip to match the bank's I/O width. Other digital peripherals include a 512-bit FIFO to prevent back pressure during stacked DRAM access, as well as power management and clock synchronization units.

4.4 3D-PATH Operation

This section presents an example of how the integration of fast-LUT and DRAM-LUT can support essential operations, using sparse matrix multiplication as a case study. Because sparse matrix-vector multiplication is widely used. Additionally, it presents an opportunity to accelerate computation and reduce storage requirements.

4.4.1 Sparse-aware Matrix Multiplication. Fig. 3 illustrates the matrix multiplication workflow in the hierarchical LUT. This work employs an outer-product mechanism to process sparse matrix multiplication, where only non-zero input vector parallel searches for matching weight elements for computation. The computation proceeds as follows: ❶ In content search mode, the non-zero input elements ($I_{0,1}$) first query the fast-LUT to identify matching row indices (IN.Index). Only entries with a valid match proceed to multiplication with input element ($W_{1,0}$ in Bank 0, $W_{1,3}$ in Bank 3). ❷ In address search mode of fast-LUT, this match result is then used to readout the corresponding DRAM-LUT address (DRAM.Addr). ❸ The command and address are transmitted through the copper pillar to DRAM. The WL of the corresponding DRAM-LUT row is activated. ❹ The input value (IN.value) serves as the column address to select the result from this row. And the OLU completes post-processing.

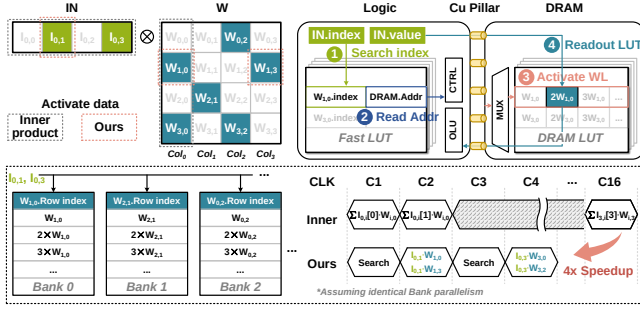


Figure 3: Sparse-aware Multiplication Workflow in 3D-PATH

To accelerate larger workloads, this work leverages a **column-interleaving** approach. Each DRAM bank computes a partial product for a different column of the output matrix, and valid input elements simultaneously identify matching weights (Bottom of Fig. 3). This parallelism is amplified by the 3D hybrid bonding architecture, which physically maps LUTs for different matrix columns to separate DRAM banks. Multiple DRAM banks can process valid inputs concurrently. The fast-LUT design separates computation into two steps, **search** and **compute**, which enables the skipping of unnecessary computations. This also reduces redundant storage. With column-interleaved mapping, each DRAM-LUT stores only non-zero weights from its corresponding matrix column, and the fast-LUT efficiently identifies matching weights during search.

4.4.2 Floating-point Matrix Multiplication. LUT-based designs offer flexibility by supporting arbitrary functions through precomputed results. However, most implementations are constrained to integer data formats, as the table size grows exponentially with higher precision FP data. To efficiently support FP multiplication in LUT, this work proposes a lossless transformation, as shown in Equ. 1. Each FP input/weight (IN/W) is composed of a sign (S), mantissa (M), and exponent (E). In FP multiplication, the signs are XORed, mantissas are multiplied, and exponents are added. The partial product is then concatenated and normalized to a standard FP output. In this design, the most computationally intensive step—the multiplication of the input mantissa and the weight—is accelerated using a LUT, which directly outputs a result in standard FP format. The remaining operations, including sign computation and exponent addition, are efficiently handled by OLU. This transformation simplifies FP multiplication to a short-width LUT access with a lightweight integer operation, substantially lowering computational complexity. As Fig. 4 shows, ① If the sliced segment consists entirely of zeros, the result is guaranteed to be zero, making the LUT lookup unnecessary. These all-zero cases are detected and skipped to avoid redundant access. ② The LUT contains precomputed products of the input mantissa and the full-precision weight, with each entry already normalized and stored in standard FP format. The results are retrieved to the logic Die. ③ Finally, the input exponent is then added to the exponent field of the LUT output, and the final sign is computed using an XOR gate. Compared to prior approaches, this method avoids time/power-consuming multipliers.

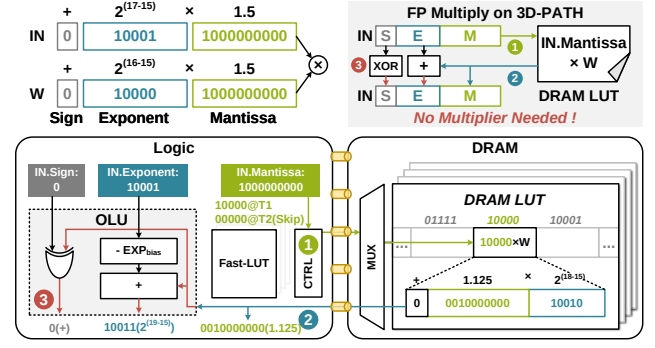


Figure 4: FP multiplication workflow on 3D-PATH

$$\begin{aligned}
 IN \times W &= (S_{IN} \oplus S_W) \cdot M_{IN} 2^{E_{IN}} \times M_W 2^{E_W} \\
 &= (S_{IN} \oplus S_W) \cdot \left\| M_{IN} \times M_W \times 2^{E_W} \right\|_{LUT} \times 2^{E_{IN}}
 \end{aligned} \quad (1)$$

FP multiplication and accumulation (MAC) further complicates LUT design due to the need for exponent pre-alignment. This step requires identifying the maximum exponent among inputs and appropriately shifting mantissas to ensure correct accumulation. Existing PIM designs address this with dedicated pre-alignment units, which can consume up to 38.5% of total compute power[70], or by performing bit-serial shifting in memory, resulting in high latency [71]. In contrast, the proposed design eliminates pre-alignment. This is enabled by the mentioned **column-interleaving** outer-product approach. Unlike prior inner-product designs, which require pre-alignment across multiple input-weight pairs for accumulation, our method assigns each DRAM bank an independent weight matrix column, performing only pointwise multiplications with incoming inputs. Since no inter-bank accumulation is needed, exponent pre-alignment is inherently avoided.

5 Thermal-aware Hardware Design

5.1 Motivation of Thermal-aware Hardware

3D-PATH benefits from hybrid bonding's key advantages: high parallelism, low access latency, and reduced power consumption. However, the thermal issue remains a fundamental challenge in hybrid bonding architectures, as inadequate heat dissipation can undermine these benefits. In this architecture, the logic Die is sandwiched between the DRAM Die and the package board, which limits its ability to dissipate heat directly through cooling methods, such as liquid or air cooling. As a result, heat accumulates during operation. According to earlier report, the thermal resistance R_{θ} of the logic Die is $1.52 \times$ higher than that of the top Die[53], causing its temperature to rise faster with the same power. To prevent overheating, the logic units must reduce their operating frequency, which ultimately compromises system performance.

Additionally, the copper pillars used for inter-Die communication also act as thermal conduits, transferring heat from the logic Die to the DRAM Die. As a result, elevated temperatures can degrade DRAM performance, particularly bandwidth, due to increased

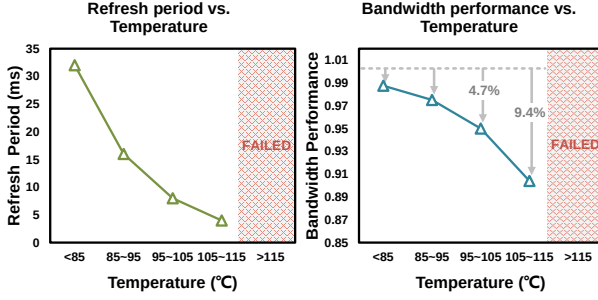


Figure 5: Refresh period and bandwidth vs. temperature

leakage currents in the storage capacitors, which reduce data retention time. To maintain data integrity, DRAM must perform periodic refreshes. However, during refresh cycles, the DRAM is unavailable for data access, leading to bandwidth loss. As shown in Fig. 5, the DRAM refresh period is temperature-dependent. Below 85°C, the refresh interval is typically 32 ms, consuming about 0.6% of total bandwidth. For every 10°C increase, the refresh period is halved; at 105°C, the refresh period drops to 8 ms, with refresh operations occupying nearly 5% of the bandwidth. When the temperature exceeds 115°C, DRAM becomes unstable and may fail to reliably retain data. These effects highlight the necessity of thermal-aware hardware to prevent performance degradation and ensure system reliability.

Thermal-aware design vs. Low power design Unlike conventional low-power designs that focus on minimizing total energy consumption, thermal-aware architectures emphasize uniform heat distribution across the chip to prevent localized hotspots that can compromise system performance and reliability. In earlier 3D systems, heat generated by the logic Die is conducted directly into the adjacent DRAM arrays. If thermal hotspots emerge in specific regions, they can impair the performance of nearby DRAM banks, reducing bandwidth utilization and introducing bottlenecks that affect the entire system, even when the average power consumption remains low [49]. Consequently, thermal-aware architecture is essential in 3D hybrid bonding systems, maintaining uniform thermal distribution and avoiding performance degradation caused by localized overheating. In this work, we propose a ‘hotspot-aware’ hardware mechanism to mitigate these potential thermal bottlenecks.

5.2 Step 1: Find the Hotspots

The first step is to detect the ‘hotspot’ components that potentially generate heat spikes. Since all modules on the logic Die use the same conductive materials, they exhibit similar thermal resistance R_θ ($\text{mm}^2 \cdot \text{K}/\text{mW}$). To estimate temperature (K) and locate hotspots, we adopt a metric of **power density** ($\text{mW}/\text{op}/\text{mm}^2$), which measures the power consumed per operation per unit area. This analysis focuses on individual operations, such as the add operation performed by a full adder. Power and area metrics are obtained through post-layout synthesis, using eight metal layers for place-and-route. Parasitic elements, including wiring and pin capacitance, are extracted to account for P&R loading. The results of power map (power/area) are computed and summarized in Table 1, providing several key observations. For example, in the Fast-LUT

Table 1: Power density of key components

Hardware	Operation	Power density ($\text{mW}/\text{op}/\text{mm}^2$)
Fast-LUT Array	Read	33.32-40.15
Fast-LUT SA	Latch results	41.76- 320.14
Fast-LUT Peripheral	Controller & Driver	67.35-81.68
Fast-LUT PC	Pre-charge Unit	38.42
Fast-LUT CL	Comparison logic	30.22-33.74
OLU Adder	Add/Sub/Comparison	99.10- 206.15
OLU Logic	Boolean Logic	9.44-11.54
OLU Shifter	Shifter	89.25-129.49
OLU Peripheral	Digital peripheral	34.12-75.71
Mesh NoC	Network-on-chip	19.78-82.65
DRAM Controller	Generate C/A	58.02-72.05
FIFO	IN/OUT FIFO R/W	37.13-39.62
Configuration	Hardware configuration	35.80

design, the power density of the array is $1.05\text{-}7.97\times$ lower than that of its sense amplifier and peripheral circuits. While the array consumes substantial power, the large area helps distribute the power consumption evenly, thereby mitigating the risk of hotspots. DRAM controller requires lower power compared to conventional 2D/2.5D DRAM designs. This is because the hybrid bonding DRAM boosts bandwidth via I/O parallelism and eliminates the need for a complex PHY layer to improve I/O data rates.

And we observe that the power density is case-dependent. For Fast-LUT SA, different current paths are activated depending on whether the input query matches or mismatches the stored data, resulting in varying current flow along the accumulation line and fluctuating power consumption. As a result, the power consumption of the SA changes. Similarly, the power density of the adder depends on the data distribution. During comparison or subtraction, signed data toggles more frequently, especially values near zero in two’s complement (2C) representation. This increases switching activity and higher power use. To address these variations, thermal-aware design aims to constrain the power density of such components toward their lower bound. In this work, both the Fast-LUT SA and the OLU adder are optimized to mitigate potential hotspots.

5.3 Step 2: Eliminate the Hotspots

Assuming similar thermal resistance across on-chip components, those with higher and more variable power density are more prone to forming ‘hotspots’. Power density analysis presents that the OLU Adder and the Fast-LUT SA exhibit the highest localized power densities under certain workloads. To address this, both components are optimized to reduce their power density, helping maintain a more uniform temperature profile across the chip.

5.3.1 Sign-magnitude Low Toggling Adder. The OLU adder exhibits high power density primarily due to frequent bit toggling during arithmetic operations. In particular, adders based on 2C representation tend to experience significant switching activity, as the least significant bits (LSBs) are likely to toggle during addition, subtraction, or comparison, particularly for values near zero. To mitigate this, this work aims to reduce switching activity during computation. An effective approach is to employ number representations that naturally limit bit transitions in typical arithmetic operations.

The sign-magnitude (SM) representation is chosen rather than 2C for the adder design, primarily to minimize power consumption during bitwise operations. In the 2C format, representing negative values involves bit inversion and an added one, which leads to frequent toggling in the LSBs. In contrast, SM representation handles sign inversion by flipping only the sign bit, significantly lowering bit transitions. This reduced switching activity directly contributes to lower dynamic power. Prior analysis shows that SM representation improves energy efficiency by 35% for uniformly distributed inputs and up to 67% for normally distributed data (with mean $\mu = 0$ and standard deviation $\sigma = 25$)[3]. The toggling rate under uniform distribution is formally derived in Equ. 3 and Equ. 4, where N represents bit width.

$$\text{Toggle} = \frac{1 * 0.5 + (N - 1) * 0}{N} = \frac{0.5}{N}, SM \rightarrow \text{Operands} \quad (3)$$

$$\text{Toggle} = \frac{N * 0.5}{N} = 0.5, 2C \rightarrow \text{Operands} \quad (4)$$

Although the SM format effectively reduces switching activity and improves energy efficiency, it introduces area overhead due to the additional logic for sign handling. In particular, when operands have opposite signs, their absolute values must be compared to determine the result's sign, necessitating a comparator that increases circuit complexity and area. This overhead can diminish the energy gains achieved through reduced bit toggling. To address this, we implement a dual-adder tree architecture that preserves the energy efficiency of SM format while minimizing the associated area cost.

In the dual-adder tree architecture, input elements are separated into two lanes based on their sign bits (Positive/Negative), and routed to two independent adder trees. Each tree processes unsigned values, eliminating the need for sign comparison logic. After computing partial sums, the outputs of the two trees are combined using a subtractor to produce the final result. Moreover, the SM format further simplifies floating-point accumulation by reducing bit toggling during alignment and sign determination. As a result, the dual-adder tree design achieves an efficient trade-off between power density and hardware complexity.

5.3.2 Mismatch-aware Self-throttling SA. This work introduces a fast-LUT design in the logic Die to enable single-cycle DRAM-LUT search, but it also raises thermal concerns due to shortcut current paths during mismatches. Specifically, when the stored data does not match the input query, a short-circuit path forms between the power supply and ground, driving current through the accumulation line and generating localized hotspots. To address this, we observe that mismatches dominate power consumption and propose a self-throttling sense amplifier that employs early gating to terminate current flow during mismatches.

Conventional SA design focuses on reducing the voltage swing of the accumulation line, but it limits the functioning range of the circuits and potentially deteriorates accuracy under PVT variation [33, 34]. Alternatively, selective-precharge schemes reduce power by precharging only parts of the accumulation line based on predictions, but they require complex controller [50, 60]. **A key observation** is that each search typically activates only a small number of DRAM-LUT rows, meaning that most fast-LUT entries

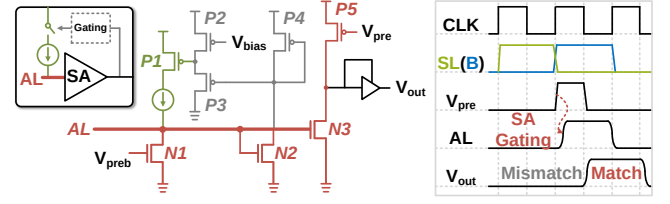


Figure 6: Circuit scheme of self-throttling sense amplifier

result in mismatches. These frequent mismatches dominate power consumption and are the primary contributors to thermal hotspots.

To mitigate hotspot formation in the sense amplifier, we propose a self-throttling design that uses early gating to terminate current sensing upon detecting a mismatch. The operation of the self-throttling circuit is illustrated in Fig. 6. Initially, the pre-charge signal sets the accumulation line to a high voltage state. During a match, this state is maintained; however, a mismatch causes the line voltage to drop, triggering the throttling mechanism. Specifically, the gate of N2 is pulled low, turning it off and raising the voltage at its drain. This action deactivates transistors P3 and P4, breaking the feedback loop and preventing activation of the current source P1. As a result, the throttling technique significantly reduces current consumption, saving up to $11.9 \times$ the power compared to a conventional SA. The average current is limited to $0.395 \mu A$. This mechanism lowers heat generation during mismatches.

6 Methodology

Environment Setup: To evaluate the performance of the 3D-PATH design, we prototyped the system in two stages, leveraging the hybrid bonding technique for independent fabrication of the logic and DRAM Dies. In the first stage, the logic Die is designed using a standard logic design flow, with Verilog HDL implementation. The fast-LUT circuit design is customized, following a mixed-signal design approach. Specifically, the schematic of the bit cell and self-throttling sense amplifier is designed using Virtuoso, and its functionality is verified via Spectre simulations. The netlist, SDC, and db generated from this schematic is merged with digital peripherals. The layout for the logic unit is realized in TSMC's 28nm 1P8M CMOS technology. Post-layout synthesis includes pin and wiring timing constraints captured in the SDC file, with the standard cell library linked to the fast-LUT, DRAM, and TSV db files. Power consumption for the logic unit is estimated via gate-level simulations using the PTPX tool, leveraging post-P&R netlists and SPEF files to clarify parasitic RC effects. In the second stage, the DRAM Die is evaluated using a vendor-supplied behavioral model that reflects its timing, refresh behavior, and power characteristics. The integration of both logic and DRAM follows a co-simulation framework consistent with prior work[35, 44].

To assess the thermal behavior of hotspot-aware hardware, a thermal model is developed using ANSYS Fluent. The model inputs include the thickness and power map of the DRAM, the hybrid bonding netlist and dimensions, the logic Die power map, the TSV distribution for the logic silicon substrate, as well as environmental factors such as heat transfer coefficients, ambient temperature ($45^\circ C$), and heat flux. The configuration is shown in Table. 2.

Baseline: For the baseline comparison, the same models are run on a 32GB Nvidia V100 GPU paired with an Intel Xeon Platinum CPU. Additionally, we reference the earlier LUT-based design pLUTo, which incorporates a 3D-stacked memory configuration[17]. Specifically, pLUTo uses a hybrid memory cube architecture with a 128KB buffer stacked on top of the memory.

Hybrid bonding Baseline: To quantitatively analyze the benefits of LUT-based paradigm, we implement an analytical baseline model (3D-base). In this baseline, the 3D-stacked DRAM functions solely as a memory component, while the fast-LUT is replaced with traditional digital logic and on-chip buffers. The baseline's compute throughput is derived from the storage density ($1.42 - 1.56\text{Mb/mm}^2$) and computing density ($0.19 - 0.76\text{TOPs/mm}^2$) of the post-layout synthesis result of the 3D-PATH. The compute-to-storage ratio is based on prior 3D design studies [73]. Thus, the specification of 3D-base logic Die is 88.2 TOPs INT8 and 42.5 TFLOPs FP16 with 126 MB on-chip memory. The DRAM retains the same 8GB storage capacity and 16.38TB/s bandwidth as used in 3D-PATH.

Benchmark: To demonstrate performance improvements over prior LUT-based PIM designs, we evaluate a comprehensive arithmetic benchmark suite introduced in previous work [17]. It includes Bit Counting and Bucket Sorting (statistical analysis), RGB2Gray (image processing), Huffman Coding and DCT Transform (data compression) [2], Fibonacci Sequence and Vector Addition (arithmetic operations). To further assess the system's capability in emerging AI workloads, we include a diverse set of neural network models: Convolution network ResNet-50 [26], MobileNet v3-large [27], LSS for autonomous driving [57], language model BERT-large [12], multimodal models Vision Transformer (ViT)-large [14], and Swin Transformer [47]. We also evaluate large language models (LLMs), such as Bloom 1.1B [61] and OPT 1.3B [76]. The models are evaluated using representative datasets: BERT with a sequence length of 512 is tested on the SQuAD dataset [58]; ResNet-50, MobileNet-v3, ViT, and SwinT are evaluated on the ImageNetv2-1k dataset [36]; LSS is tested on autonomous driving datasets nuScenes [6]; and LLMs are assessed on the Wikilingua dataset [37].

7 Experiment Results

7.1 Thermal Analysis

As discussed earlier, the temperature of the DRAM directly impacts the refresh period. Below 85°C , the refresh period reaches its maximum value of 32 ms. Therefore, the ideal operating temperature for the DRAM is set at 85°C . The initial thermal analysis of this work is presented in Fig. 7. Each logic bank, paired with a corresponding 2 MB DRAM bank, consumes 96 mW of power. The system employs an air-cooling setup with a heatsink and thermal dissipation cover. With a 2U radiator height and wind speeds of 5 m/s between the radiator fins, the maximum temperatures of the DRAM and logic units are 100.5°C and 102.3°C , respectively. The DRAM temperature exceeds the target by 18.23%, resulting in a 5% decrease in bandwidth utilization. When using a 4U radiator and increasing airflow to 10 m/s, the maximum temperatures of the DRAM and logic are reduced to 95.2°C and 97.0°C , respectively. However, the DRAM temperature still exceeds the ideal by 12%, requiring an 8 ms refresh period and resulting in a 2.4% bandwidth loss.

Table 2: Design specification of DRAM and logic Die.

DRAM Die	Specification
DRAM Die	8GB, 4096 Banks, 256 Bank groups 2MB/bank (w/ ECC 64 correct 1)
Bank	4096 rows $\times$ 4096 cols, $398\text{ }\mu\text{m} \times 348\text{ }\mu\text{m}$ (Cu Hybrid Bonding)
Data rate	64 I/O/bank, 500MHz $\times 64$ burst 16.38TB/s Bandwidth/Chip
Timing constraint	tRCD = 16.4ns, tCL = 1.7ns, tRAS = 33.6ns, tRP = 11.7ns, tRC = 45.3ns
Operating voltage	VDD (main voltage) $1.2 \pm 0.02\text{V}$ VCC (circuit) $1.1 \pm 0.06\text{V}$ VEQ 1.4V Vios (IO voltage) $0.9 \pm 0.05\text{V}$ Vlogic 0.9V
Logic Die	Specification
Bank Area	$398\text{ }\mu\text{m} \times 348\text{ }\mu\text{m}$ (28nm CMOS)
Fast-LUT	32Kb/group, 0.0495 mm^2 (28nm CMOS)
OLU	Dual-adder tree, Boolean logic Normalization unit Exception & Rounding unit
Peripheral	Inter-bank group NoC (Mesh) 64b-burst 512b FIFO, Power & Clock
Frequency	1GHz
Operating voltage	VDD (main voltage) $0.9 \pm 0.02\text{V}$ VSA (sense amplifier) $1.0 \pm 0.05\text{V}$ Vios (IO voltage) $0.9 \pm 0.05\text{V}$
Hybrid Bonding	Specification
Cu Bonding	$2.25\text{ }\mu\text{m}$ pitch, $2.3\text{ }\mu\text{m}$ length
TSV	X-axis $160\text{ }\mu\text{m}$ pitch, Y-axis $156\text{ }\mu\text{m}$ pitch
Heat Sink	Specification
Dissipation	Vapor Chamber + Heat Pipe
Sink dimension	$120\text{ mm} \times 98.4\text{ mm} \times 2\text{U}/4\text{U}$
Wind speed	5 m/s, 10 m/s
Environment	Ambient temperature: 45°C

To meet the target DRAM temperature, we further investigate the allowable power consumption of the logic Die under various heat dissipation techniques. Fig. 8 illustrates the relationship between logic power consumption and the temperatures of both the logic and DRAM Dies. For a fin wind speed of 5 m/s, when the logic power consumption is reduced to 14.06 mW, the temperature of the logic Die decreases to 85.8°C , and the DRAM Die temperature drops to 84.8°C , thus meeting the target DRAM temperature. However, this requires a 40.01% reduction in allowable logic power. Increasing the wind speed to 10 m/s improves thermal dissipation. The DRAM temperature decreases to 84.9°C when the logic power is reduced to 18.75 mW, meaning a 20.00% reduction in the allowed logic power.

To tackle thermal challenges, we introduce a hotspot-aware hardware design that targets components with high power density. The SA is optimized with a self-gating mechanism upon detecting mismatches, which is based on the observation that mismatches dominate content-search operations. In representative workloads, the average match rate across 60 fast-LUT iterations is less than 3.2%. We evaluate the impact of the self-throttling technique under various temperature conditions, as shown in Fig. 9. The self-throttling

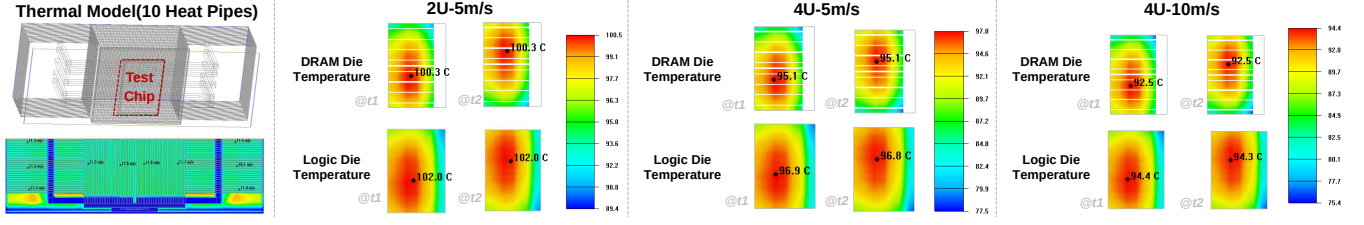


Figure 7: Thermal model; Temperature of logic and DRAM Die results under different cooling systems

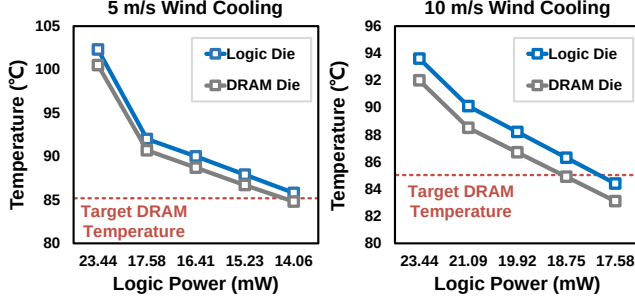


Figure 8: Temperature of logic and DRAM Die correspond to the logic bank power

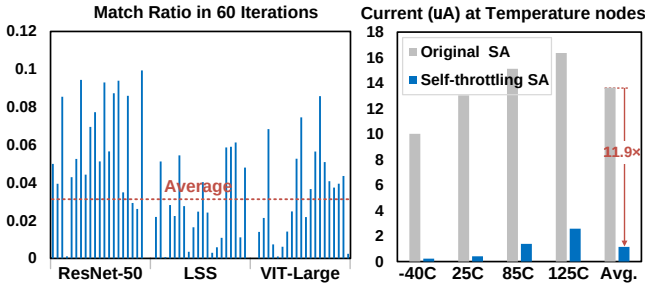


Figure 9: Match ratio of sense amplifier in representative applications; Current saving of self-throttling sense amplifier under different temperature nodes

SA achieves a maximum reduction of 43.4 $\times$ in sense current. Although this saving decreases with increasing temperature due to higher leakage currents, it still delivers a 6.34 $\times$ current saving at 125°C, resulting in an average power saving of 11.9 $\times$ across all tested temperature points.

Additionally, this work uses the SM format to reduce the toggle rate and a dual-adder tree efficiently supports this representation. As shown in Fig. 10, the SM format reduces toggle rates by 27.9% for integer operations and 16.6% for floating-point operations compared to 2C. Regarding the power consumption, the design results in a 40.5% power saving at room temperature with 50% sign inversion, with an average saving of 36.4% across all temperature points. For floating-point operations, the power reduction is less significant due to additional steps like normalization, alignment, and exception handling, which are not optimized by the dual-adder. Nevertheless, the SM-based design still yields a 13.2% power saving.

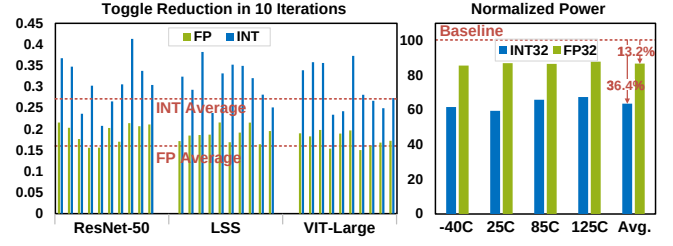


Figure 10: Toggle reduction of SM compared with 2C representation; Normalized power, power of 2C is 100 %.

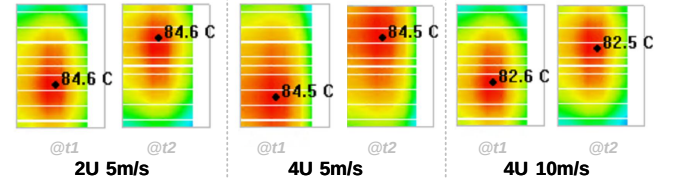


Figure 11: Optimization temperature performance

The thermal impact of the SA and adder optimizations is illustrated in Fig. 11. With these enhancements, the DRAM temperature consistently stays below the target threshold of 85°C across all combinations of fin speed and radiator height. At a fin speed of 5 m/s, the logic Die temperature ranges from 85.9°C to 94.8°C. When the fin speed is increased to 10 m/s, the temperature further decreases to a range of 81.6°C to 87.7°C. The hotspot-aware hardware results in a 17.1% reduction in logic Die power consumption, which is lower than the required power reduction to meet the target DRAM temperature. However, the highest temperature spot in the DRAM is limited below the target, which can be attributed to the hotspot-aware hardware's ability to mitigate hotspots and ensure more uniform heat dissipation. Despite not limiting the logic power below the target, the optimizations still successfully maintain the DRAM temperature below the critical threshold of 85°C.

7.2 Circuit Analysis

To enable efficient redundancy-aware computing, 3D-PATH incorporates a customized fast-LUT that facilitates hierarchical LUT-based computation. The fast-LUT components are implemented from gate-level circuit to manual layout. The following experiments evaluate the functionality and performance of the circuit design.

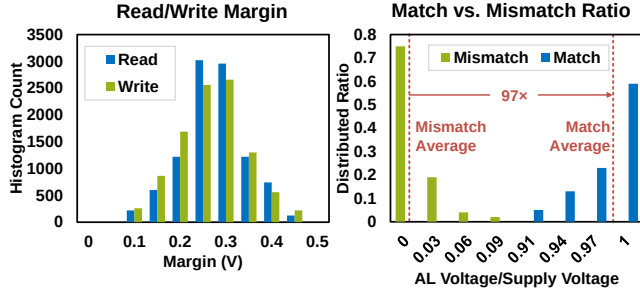


Figure 12: Read/write margin of fast-LUT cell; Match/Mismatch ratio of Fast-LUT cell

7.2.1 Fast-LUT cell. The fast-LUT cell is based on a modified SRAM 6T cell, which is derived from an industry design[8]. The schematic is implemented using TSMC's 28nm process. To assess its robustness under process, voltage, and temperature (PVT) variations, post-layout simulations are performed, as shown in Fig. 12. First, a Monte Carlo analysis with 10,000 iterations is conducted to evaluate the read and write margins of the fast-LUT cell, which are critical for determining the cell's stability. The read margin is 275 mV, while the write margin is 268 mV, both exceeding 30% of the supply voltage, confirming robust noise immunity and reliable data retention. Second, the fast-LUT's ability to distinguish between match and mismatch scenarios is analyzed. In the mismatch case, the fast-LUT discharges the accumulation line, while in the match scenario, the AL is maintained at a high state. Experimental results show that the average mismatch voltage is 0.01 VDD, while the match voltage is 0.98 VDD, achieving a match/mismatch ratio greater than 97 $\times$. The results validate its efficiency in distinguishing two scenarios.

7.2.2 Self-throttling Sense Amplifier. The experiment presented in Fig. 13 evaluates SA sensing latency. The shortest sensing latency is achieved at the minimum bias voltage, where the current source is fully activated, rapidly driving the voltage to the qualified threshold within 0.12 ns. As the bias voltage increases, the current source gradually shuts off, leading to longer sensing times. At a bias voltage of 0.4 V, the sensing latency reaches 0.45 ns. Beyond this point, the limited current source fails to drive the accumulation line to the target threshold voltage, particularly at lower temperature nodes. A key trade-off in this design is that faster sensing typically incurs higher power consumption due to increased current draw. At room temperature, the minimum sensing power is observed at 0.4 V Vbias, measuring 0.55 fJ/bit, while the maximum sensing power reaches 1.47 fJ/bit at 0.1 V. This ability allows users to configure the bias voltage based on specific application requirements. Given the logic Die operates at a target frequency of 1GHz, the latter half of each clock cycle is reserved for current sensing. To ensure timing safety, Vbias is constrained between 0 and 0.3V, where the corresponding sensing delay stays within the safe range of 0.12-0.32ns. For the subsequent experiments, the bias voltage is fixed at 0.2 V.

7.3 Design Space Exploration and Overhead

7.3.1 Memory array organization & overhead. The lowest granularity of DRAM organization is DRAM MAT, which consists of

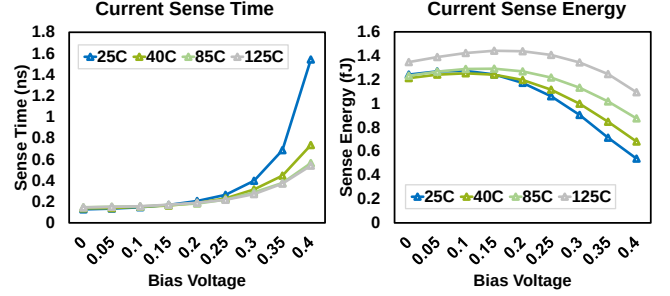


Figure 13: Self-throttling SA sense energy and time

2048 wordline $\times$ 4096 bitline. Two MATs share a common set of sense amplifiers and together form a 2MB DRAM bank. Column circuitry selects 64b from row buffer and delivers it to the logic Die through pillar. The MAT size is chosen to balance robustness and area efficiency. Specifically, larger DRAM MAT structure induces severe signal deterioration along long wiring, while smaller MATs incur fixed overheads, such as ECC, that are difficult to amortize.

The organization of DRAM-LUT entries is determined based on the underlying DRAM MAT structure. The DRAM bank has a 4Kb page size and supports open page policy. LUT entries are preferentially allocated along the row direction. As a result, the number of entries multiplies the entry size is constrained to 4096 bits, matching the row width of the DRAM bank. The DRAM-LUT update process also benefits from the high bandwidth enabled by the hybrid bonding. Each bank can be independently accessed via pillars in 1 cycle, and the short path ensures fixed latency updates.

7.3.2 Memory controller design & overhead. In the hybrid bonding design, the DRAM remains intact, while the memory controller is integrated into the logic Die. The distributed controllers are necessary for generating physical addresses, issuing instruction sequences, and managing data interfaces. But the increased parallelism leads to a higher controller overhead. To address this, each bank group incorporates a centralized memory controller that consolidates data I/O from 16 physically adjacent banks. At the full-chip level, 256 parallel memory controllers occupy 17% of the logic Die area.

7.3.3 Integration I/O & overhead. The upper limit number of pillars is constrained by its minimum pitch size. These pillars serve multiple functions, including signal I/O, power delivery, and mechanical support, where redundant pillars help mitigate chip warpage. The required number of signal I/Os is determined by the memory array. This design adopts a $\times 64$ burst, resulting in 64b R/W data I/O, along with 12b column address and 6b row address bus. Additional signal I/Os include 8b command I/O, 23b TRIM for user set parameter, 8b for BIST. The copper pillars incur no extra area overhead, as they only connect the top metal layers of two Dies without impacting intra-Die place & route. Signal I/O is synchronized with RDQS/RDQSB (Fig. 1) and interacts with logic Die through a digital protocol. The interface does not rely on complex PHY to boost frequency, thus the I/O frequency is limited at 500MHz. But with high I/O parallelism, the total bandwidth surpasses that of 2D/2.5D approach.

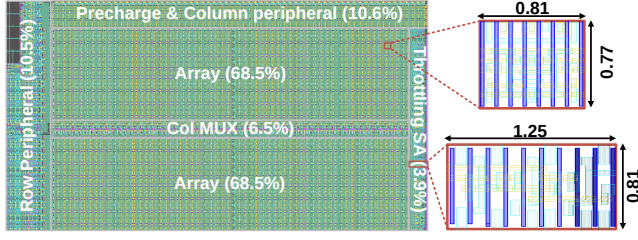


Figure 14: Floorplan of fast-LUT, cell and self-throttling SA

7.3.4 Power supply & overhead. The power supply requirement is concluded in Table. 2. The power of logic Die is supplied through the path from package board, passes through TSV, and reaches the power distribution network on the logic Die. For the DRAM Die, power is also delivered via this same path, but further requires vertical transfer through copper pillars from the logic Die. Therefore, the power delivery of the entire chip relies on TSV. Each TSV requires an area of $25 \mu\text{m}^2$ and a 2:1 redundancy. TSVs also introduce placement constraints, as design rules prohibit active routing or logic placement within a deadspace. To meet power requirements and avoid IR drop, the entire chip adopts an average TSV pitch of $88.7 \mu\text{m}$, with each TSV supporting a maximum current of 36 mA. The total area overhead introduced by TSVs is 0.4%.

7.3.5 Fast-LUT design & overhead. We present a fast-LUT design to enable efficient sparse-aware computing. The floorplan of the fast-LUT is shown in Fig. 14. The basic 9T cell achieves an area of $0.62 \mu\text{m}^2$, improving area efficiency by 1.48-4.03 $\times$ than conventional CAM cell design [13, 42, 45]. Each 32 Kb fast-LUT occupies an area of 0.0495 mm^2 . The area of Fast-LUT is smaller than the corresponding DRAM-LUT to accommodate the other digital peripheral. Notably, the fast-LUT can also function as a conventional SRAM. When compared to a 32 Kb SRAM generated by a memory compiler, the fast-LUT design exhibits a 30.1% area overhead.

7.3.6 Thermal-aware hardware & overhead. To address thermal challenges, this work integrates a self-throttling SA and a dual-adder tree. The self-throttling SA occupies only 3.9% of the fast-LUT area, and aligns with the column pitch of the storage array, avoiding additional placement and routing. The dual adder tree splits the initial adder into parallel pos/neg adders, which introduces less than 0.1% of the total logic Die area.

7.4 Overall Architecture Performance

Fig. 15 presents a comparison of throughput performance between 3D-PATH, 3D-base, the baseline GPU, and the LUT-PIM pLUTo on arithmetic applications. 3D-PATH consistently outperforms the GPU, achieving an average throughput improvement of 7.78 $\times$. This significant gain is primarily due to the bank-level parallelism, along with reduced access latency enabled by the hybrid bonding technique. In comparison to pLUTo PIM, 3D-PATH achieves a constant throughput of 16.38 TOPS, yielding an average performance improvement of $2.27 \times$ in arithmetic benchmarks. While pLUTo also leverages DRAM array parallelism, it requires iterative searches, which limits its efficiency. In contrast, 3D-PATH benefits from the high parallelism and burst capabilities provided by hybrid bonding,

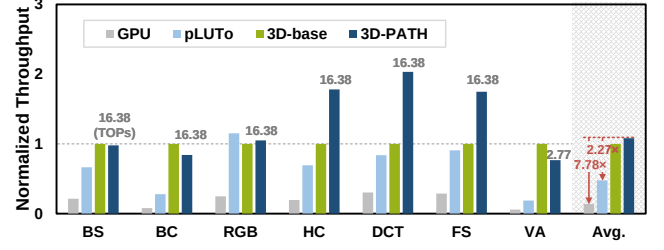


Figure 15: Throughput performance (TOPS): Results are normalized to the hybrid bonding baseline. Absolute values of 3D-PATH are also labeled.

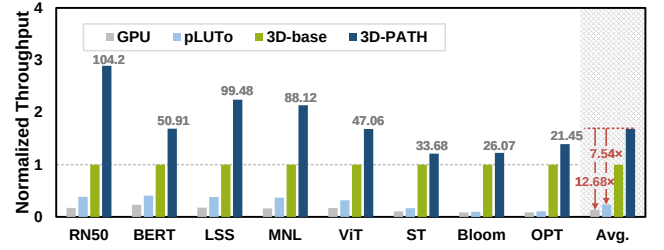


Figure 16: Throughput performance (TOPS): Results are normalized to the hybrid bonding baseline. Absolute values of 3D-PATH are also labeled.

which significantly boosts LUT throughput. The 3D-base also shows $2.10 \times$ improvement over pLUTo due to the high bandwidth. On specific applications such as Bit Counting (BC), its parallel digital array enables it to outperform 3D-PATH by 1.26 $\times$. But 3D-base suffers from long latency when handling complex operations, which can be easily resolved by 3D-PATH. On average, 3D-PATH improves performance by 1.09 $\times$ than 3D-base.

Fig. 16 presents the throughput performance on neural network benchmarks. A wide range of matrix multiplication tasks is included, from 64×64 (small-scale) to 1024×1024 (large-scale). Each operation (Op) is defined as an integer multiplication or accumulation. 3D-PATH achieves 58.87 TOPs, 12.68 $\times$ higher than GPU, 7.13 $\times$ higher than pLUTo, 1.68 $\times$ higher than 3D-base. This gain is primarily due to the hierarchical LUT design, which efficiently locates non-zero elements and skips redundant computations. Models with higher sparsity show more throughput improvements. For example, 3D-PATH outperforms pLUTo and 3D-base by 7.53 $\times$ and 2.89 $\times$ respectively on ResNet-50, and by 5.89 $\times$ and 2.24 $\times$ on the LSS model. The area efficiency of 3D-PATH can be further acquired by dividing the chip area. The area efficiency of 3D-PATH is 103.77 GOPs/ mm^2 on average. By stacking two wafers together, the hybrid bonding scheme naturally enhances the area efficiency.

Fig. 17 shows the energy efficiency comparison between 3D-PATH and the other implementations on arithmetic applications. 3D-PATH optimizes power consumption in two key ways. First, energy for DRAM access is reduced by minimizing access distance using copper pillars. Specifically, the energy per bit for DIMM is 12 pJ/bit, for HBM is 4 pJ/bit, and hybrid bonding reduces this to just 0.3 pJ/bit. Second, thermal-aware circuit optimizations further

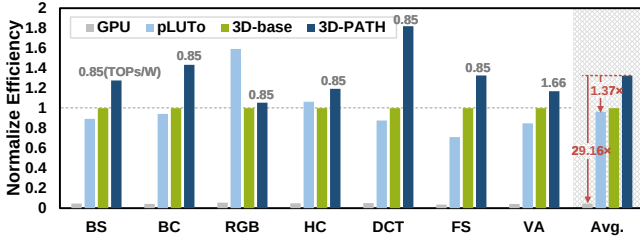


Figure 17: Energy efficiency (TOPS/W): Results are normalized to the hybrid bonding baseline. Absolute values of 3D-PATH are also labeled.

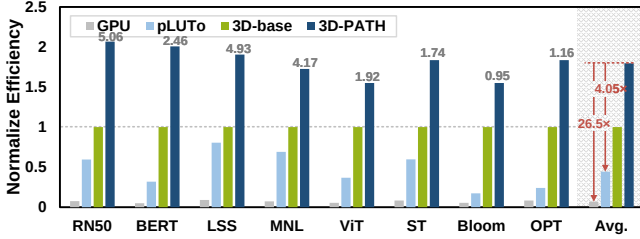


Figure 18: Energy efficiency (TOPS/W): Results are normalized to the hybrid bonding baseline. Absolute values of 3D-PATH are also labeled.

lower power consumption. When compared to the GPU baseline, 3D-PATH achieves an energy efficiency of 0.966 TOPs/W and 29.16× improvement in energy efficiency. Although pLUTo achieves high energy efficiency in vector-parallel operations through in-situ processing, it suffers from the need to duplicate LUTs, which reduces storage efficiency and incurs additional data loading overhead. The 3D-base design benefits from hybrid bonding for lower memory access energy, but its parallel MAC array leads to higher dynamic power. In comparison, 3D-PATH provides an overall energy efficiency improvement of 1.37× and 1.32× over pLUTo and 3D-base. On neural network benchmarks, 3D-PATH achieves 2.80 TOPs/W considering sparsity processing, demonstrating its advantage in energy efficiency. As Fig. 18 shows, 3D-PATH achieves 4.05× and 1.80× improvement over pLUTo and 3D-base.

Earlier LUT-based PIM designs hardly supported floating-point operations due to the exponential growth in LUT size. The 3D-PATH addresses this limitation by incorporating representation transformation and column-interleaving techniques, enabling efficient FP computation without compromising accuracy. To validate the efficiency of FP operation, Fig. 19 compares the performance of 3D-PATH with GPU baseline and 3D-base on neural network benchmarks, utilizing FP16 data format. On average, 3D-PATH achieves 27.1 TFLOPs, offering a 4.29× improvement in throughput over the GPU, 1.71× over 3D-base. This performance gain is mainly attributed to the exploitation of model sparsity. Compared with 3D-base, 3D-PATH delivers a 2.53× throughput improvement on ResNet-50 and a 3.12× improvement on LSS, and a 1.75× gain on the OPT model. In terms of energy efficiency (Fig. 20), 3D-PATH outperforms GPUs by leveraging the low access power of hybrid-bonding

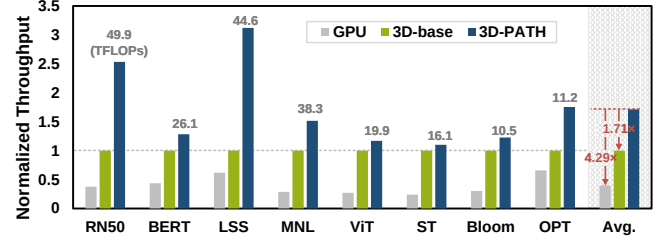


Figure 19: Floating point throughput performance (TFLOPs): Results are normalized to the hybrid bonding baseline. Absolute values of 3D-PATH are also labeled.

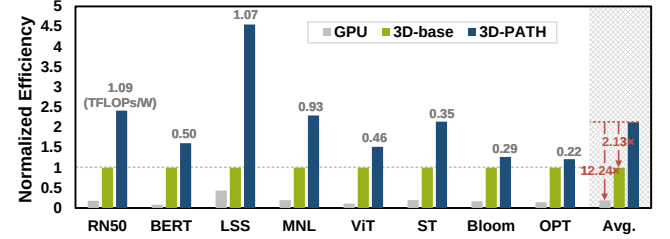


Figure 20: Floating point energy efficiency (TFLOPs/W): Results are normalized to the hybrid bonding baseline. Absolute values of 3D-PATH are also labeled.

Table 3: Comparison of 3D-PATH vs. prior DRAM-PIMs. All performances are evaluated under identical parallelism

	Ambit[63]	SIMDRAM[25]	LAcc[11]	DRISA[43]	pLUTo[17]	3D-PATH
Capacity	8GB	8GB	8GB	2GB	8GB	8GB
Technology	2D-DRAM	2D-DRAM	2D-DRAM	2D-DRAM	2D/HMC	3D-DRAM
Power	5.3	5.3	5.3	98.0	11	76.8
NOT (ns)	135	135	135	207.6	105.0	36
AND (ns)	270	270	270	415.2	165.0	36
OR (ns)	270	270	270	415.2	165.0	36
XOR (ns)	585	585	450	691.9	165.0	36
XNOR (ns)	585	585	450	691.9	165.0	36
Energy Efficiency*	0.8917	0.8917	1.15	0.0407	1.523	1
4-bit Addition (ns)	5081	1585.0	1142.3	1756.5	1920.0	36
4-bit Multiplication (ns)	19065	7451.0	5365.4	8250.1	1920.0	36
4-bit Bit Counting (ns)	2936	1156.0	-	6649.9	120.0	36
8-bit Bit Counting (ns)	6901	2696.0	-	13580.0	1920.0	36
Energy Efficiency	0.0756	0.1934	0.4565	0.0020	0.1309	1
FP16 Multiplication	-	-	-	-	-	72
FP32 Accumulation	-	-	-	-	-	72

DRAM. Models with lower computational intensity particularly benefit from the 3D integration. For instance, Bert demonstrates a 19.9× improvement in energy efficiency, while ViT achieves a 14.07× gain. On average, 3D-PATH enhances energy efficiency by 12.24× compared to the GPU baseline. Moreover, by eliminating the need for multipliers and alignment logic in FP operations, 3D-PATH outperforms 3D-base in energy efficiency by 2.13×.

7.5 Comparison with Prior PIM Designs

Table. 3 provides a comparison of 3D-PATH with prior PIM designs using metrics and normalization rules in earlier works [17]. These metrics encompass both Boolean logic and arithmetic operations. Ambit achieves in-situ computing through a triple-row

activation approach [63]. SIMDRAM enhances PIM functionality by introducing a general-purpose PIM framework [25]. LAcc implements a DRAM-based PIM accelerator to support fast and accurate multiplication [11]. DRISA integrates boolean logic and additional logic within DRAM to perform complex computations [43]. pLUTo introduces a LUT-based PIM by duplicating tables and iteratively accessing content to achieve high parallelism [17]. The results indicate that 3D-PATH achieves faster operations. This improvement is due to the localization ability of fast-LUT, and the 3D architecture, which minimizes access latency.

However, for Boolean logic operations, the energy efficiency improvement of 3D-PATH is not obvious, or even lower than that of earlier PIM designs. Because prior designs, such as Ambit, leverage the physical structure of DRAM arrays to perform Boolean logic (e.g., AND/OR operations), eliminating the need for additional computation logic. In contrast, while 3D-PATH benefits from hybrid bonding integration that reduces I/O power, the logic Die introduces additional power consumption, offsetting these gains. But a key limitation of prior PIM architectures is their restricted computational flexibility. Complex arithmetic operations require iterative row activations, leading to significant overhead in both execution time and energy consumption. 3D-PATH overcomes this limitation by leveraging a high-capacity DRAM-LUT that efficiently supports sophisticated operations. Furthermore, the logic Die facilitates additional computations beyond a single DRAM-LUT. Notably, 3D-PATH is the first hybrid bonding-based LUT, and the first LUT-based PIM to support FP operations and sparsity-aware acceleration.

8 Conclusion

This work presents 3D-PATH, a hierarchical LUT-based architecture that leverages hybrid bonding to achieve high-bandwidth, fine-grained memory access and bank-level parallelism for LUT-driven computation. The logic Die integrates a fast-LUT for efficient sparse data localization, reducing redundant storage and accelerating computation. Additionally, 3D-PATH efficiently supports multiplier-free/alignment-free floating-point operations. To mitigate the 3D stacking performance loss caused by thermal challenges, we introduce hotspot-aware hardware, including a self-throttling sense amplifier and a low-toggle adder. These components dynamically adapt to workload conditions, effectively mitigating thermal hotspots and ensuring operation within safe thermal margins. Overall, 3D-PATH achieves significant improvements in energy efficiency, ranging from 1.37× to 29.16× over existing GPU and PIM designs.

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